[Method] for [Vehicle Control Task] Under [Operating Conditions]

First A. Author, Second B. Author, and Third C. Author

***Abstract*—[Write the problem, research gap, proposed mechanism, evaluation protocol, measured results, and supported conclusion here.]**

***Index Terms*—Automated vehicles, learning-based control, model predictive control, trajectory tracking.**

I. Introduction

A. Background and Motivation

[Describe the vehicle task, operating conditions, practical consequences, and unresolved technical challenge.]

B. Related Work and Research Gap

[Compare learning-based predictive models and hierarchical vehicle control approaches.][**? ?**]

[Contrast adaptive objectives, stability-aware control, and data-driven feasibility analyses; identify the exact remaining gap.][**? ? ?**]

C. Main Contributions

[State the research objective and list the verified contributions.]

- *[Methodological advance and difference from the closest prior method; identify the supporting section.]*
- *[Analytical or computational advance and its assumptions; retain only if established in this study.]*
- *[Experimental evidence that resolves the research question, including operating conditions and reproducibility.]*

The remainder of this article presents the problem formulation in Section **??** and the proposed method in Section **??**. Section **??** develops the theoretical analysis. Sections **??** and **??** describe the evaluation and discuss the results. Section **??** concludes the article.

II. System Modeling and Problem Formulation

A. Vehicle Model and Notation

[Define the vehicle dynamics, measurements, control inputs, coordinate frames, sampling period, and model parameters.]

B. Operating Domain, Uncertainty, and Assumptions

[Specify the operating domain, measurable quantities, disturbance model, actuator limits, and justified assumptions.]

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Corresponding author: [Full name]. This work was supported by [Funder] under Grant [Identifier].

C. Control Objectives and Constraints

[Formulate the control problem, tracking objectives, safety-related constraints, and admissible operating conditions.]

III. Proposed Learning-Based Predictive Control Method

A. Overall Architecture

[Describe the architecture, module interfaces, timing, and the location of the proposed technical contribution.]

B. Learning-Based Prediction Model

[Define the learned quantity and its interface with the nominal dynamics and the predictive controller.]

1) Offline Identification and Training: [Specify identification data, training procedure, hyperparameters, and the validation-only selection protocol.]

2) Online Inference and Uncertainty Handling: [Describe online inputs, uncertainty treatment, and any adaptation mechanism or frozen-model assumption.]

C. Predictive Control Formulation

[Present the complete finite-horizon optimization problem and explain the receding-horizon implementation.]

1) Objective Function: [Define the stage and terminal costs and justify their weights.]

2) Constraints and Feasibility Management: [Define admissible states and inputs, slack penalties, terminal ingredients, and the infeasibility handling policy.]

D. Implementation and Computational Complexity

[Provide a reproducible execution procedure, solver settings, computational scaling, and timeout handling.]

IV. Theoretical Analysis

A. Recursive Feasibility and Constraint Satisfaction

[State and prove the applicable feasibility property, including all assumptions and its relationship to implementation.]

B. Closed-Loop Stability or Error Bounds

[Establish the claimed closed-loop property and explain the assumptions under which it holds.]

V. Experimental Setup

A. Platforms and Operating Scenarios

[Describe the platforms, scenario matrix, parameter ranges, execution hardware, and experimental termination criteria.]

B. Data Protocol and Baseline Methods

[Specify independent data partitions, trial counts, random seeds, baseline implementations, and fair tuning conditions.]

C. Evaluation Metrics and Statistical Protocol

[Define primary and secondary metrics, aggregation rules, independent repetitions, confidence intervals, and failure reporting.]

VI. Results and Discussion

A. Prediction Accuracy and Model Validation

[Report held-out prediction performance and relate modeling errors to the downstream control task.]

B. Closed-Loop Performance and Constraint Satisfaction

[Compare the methods across the full scenario matrix, including performance, constraint violations, and unsuccessful trials.]

C. Ablation and Sensitivity Analysis

[Isolate the contributions of the proposed components and report sensitivity to the parameters that affect deployment.]

D. Real-Time Performance and Experimental Validation

[Report implementation timing, missed deadlines, and the available hardware-in-the-loop or real-vehicle validation.]

E. Limitations and Generalization

[Discuss the observed limitations, unresolved failure modes, generalization evidence, and practical deployment implications.]

VII. Conclusion

[Summarize the verified contribution, the most informative measured findings, the supported scope, and a concrete next research step.]

Acknowledgment

[Insert verified acknowledgments and any applicable disclosure.]

References

[1] Y. Li, Z. Chen, T. Wang, X. Zeng, and Z. Yin, "Data-driven hierarchical model predictive control for automated overtaking maneuver via Gaussian process regression," *IEEE Trans. Veh. Technol.*, vol. 74, no. 1, pp. 263–278, Jan. 2025, doi: 10.1109/TVT.2024.3453170.

[2] G. Chen, X. Zhao, Z. Gao, Z. Gao, X. Wang, and M. Hua, "Experimental evaluation of trajectory tracking with nonlinear model predictive control enhanced by physics-informed neural network," *IEEE Trans. Veh. Technol.*, vol. 75, no. 6, pp. 8928–8939, Jun. 2026, doi: 10.1109/TVT.2025.3635878.

[3] J. Bae, J. Lim, B. Suh, J. Lee, K. Ryu, J. Kim, and J. Choi, "Learning expert-level racing strategies via scheduled cost functions in model predictive control," *IEEE Trans. Intell. Veh.*, vol. 10, no. 6, pp. 3827–3840, Jun. 2025, doi: 10.1109/TIV.2024.3465598.

[4] Z. Zheng, S. Wang, X. Zhao, Q. Yu, H. Zhang, Y. Lv, and J. Tian, "Adaptive model predictive control of four-wheel drive intelligent electric vehicles based on stability probability under extreme braking conditions," *IEEE Trans. Intell. Veh.*, vol. 11, no. 5, pp. 618–628, May 2026, doi: 10.1109/TIV.2024.3398129.

[5] H. Yang, Y. He, C. Mu, and P. Li, "Data-driven model predictive longitudinal control for autonomous vehicles with unknown chassis," *IEEE Trans. Veh. Technol.*, vol. 74, no. 8, pp. 11732–11742, Aug. 2025, doi: 10.1109/TVT.2025.3552933.

First A. Author *[Insert verified education, current position, and research interests. Repeat for every author.]*

Second B. Author *[Insert this author's verified biography.]*

Third C. Author *[Insert this author's verified biography.]*